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Heterogeneous spillovers of housing credit policy ☆

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ABSTRACT

We study the spillovers from government intervention in the mortgage market on households' consumption. Expansionary credit policy increases the consumption of homeowners with mortgage debt significantly, while the consumption response of homeowners without mortgage debt is small and insignificant. Non-homeowners also increase their consumption but less than mortgagors. We also find heterogeneous responses of households of different ages. We explain these facts through a life-cycle model with incomplete markets and endogenous housing choice. Downward pressure on the credit and interest rates creates extra wealth for the mortgagors via the refinancing channel.

1. Introduction

Residential mortgage debt in the US accounts for a majority of household borrowing.[1] The US mortgage market is also distinctive since the US government actively engages in mortgage market operations via several Government-Sponsored Enterprises (GSEs) and Governmental Agencies. While not allowed any direct lending to the households, at the peak of their activity, these agencies held almost 20% of the overall mortgage debt in the economy.[2] It is well-documented (see Fieldhouse et al., 2018) that portfolio purchases of the GSEs boost mortgage lending, lower mortgage rates, and influence prices on other financial markets. In this paper, we study how this economic activity affects the most significant component of GDP, household consumption. We find that portfolio purchases by the GSEs generate a significant response of households' consumption. We show that households' financial position is crucial in understanding the spillovers from these purchases

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1 For instance, in 2017, the share of residential mortgage debt was about 80% of total household borrowing.

2 GSEs were established by US Congress to support secondary mortgage markets by buying and securitizing loans from primary mortgage lenders.

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to private consumption. First, using the household-level consumption data, we empirically document and quantify the response of individuals' consumption to activity in secondary mortgage markets. Second, through the lens of a life-cycle model with incomplete markets and endogenous housing choice, we explain this empirical evidence and analyze the transmission mechanism through which mortgage market activities affect consumption.

In our empirical exercise, we explore the link between an increase in agency purchases (what we call expansionary credit policy changes) and an increase in households expenditure. To do so, we proxy households' financial position with housing tenure status. We show that following an expansionary policy change to the secondary mortgage markets, homeowners with mortgage debt increase their spending substantially. In contrast, homeowners without mortgage debt do not react to policy change. Non-homeowners also increase their consumption but less than mortgagors.

Moreover, we analyze how the same policy affects households of different ages. Complementary to the results on housing tenure status, we find that young and middle-aged households are most exposed to the activity in secondary mortgage markets and increase consumption significantly as a result of the policy.

In identifying expansionary credit policy changes, we focus on changes through exogenous governmental intervention in the mortgage markets via various federal housing agencies. For the most part, credit policy changes are a reaction to business cycle conditions (the most recent QE3 being the prime example). To analyze the consumption response to any of these policy changes, it is, therefore, essential to isolate the policy changes that are orthogonal to the business cycle (such as long-term objectives of increasing homeownership). We combine the exogenous non-cyclically motivated events from Fieldhouse and Mertens (2017) with mortgage purchases of the two largest federal housing agencies (Fannie Mae and Freddie Mac). We then use the former as an instrument in regressions of households' consumption on measures of agency purchase activity. We measure consumption using household-level data from the Consumer Expenditure Survey. If credit market interventions were neutral (Meltzer, 1974; Greenspan, 2005; Lehnert et al., 2008; Fieldhouse et al., 2018), an increase in agency purchases should have little impact on private consumption. We find empirically that credit policy changes lead to an increase in private consumption, more so for mortgagors and younger households.

In our theoretical exercise, we use a structural model to identify the transmission mechanism we found in our reduced-form analysis. We present a life-cycle model with incomplete markets and endogenous housing choice in the vein of Kaplan et al. (2020); Favilukis et al. (2017); Sommer and Sullivan (2018); Wong (2021). We model the credit policy change experiment by replicating the aggregate macroeconomic effect of mortgage market interventions documented in Fieldhouse et al. (2018). In particular, we focus on the change in both interest and mortgage rates as well as on the change in the spread between the two, as documented in Fieldhouse et al. (2018).

Our first finding is that lower mortgage rates imply lower mortgage payments for the mortgagors and a rise in long-term permanent income for this group. In terms of mechanisms in the model, we identify access to refinancing as a crucial transmission mechanism for expansionary credit policy. These results are in line with both the recent microeconomic evidence (Wong (2021) emphasizes the role of refinancing for transmission of monetary policy to consumption) as well as macroeconomic one (following a credit policy shock Fieldhouse et al. (2018) show an increase of mortgage originations due to refinancing). Since the opportunity cost of saving goes down when the interest rates drop as well - mortgagors consume this extra income instead of saving. The results we find are also in line with Cloyne et al. (2020), who argue that the behavior of mortgagors resembles that of wealthy hand-to-mouth households and empirically document a similar response of individual consumption to expansionary monetary policy shock. Indeed, in the model, mortgagors hold little liquid wealth, outstanding mortgage debt, and illiquid assets in the form of the house.

We then analyze the response of other types of households: renters and outright homeowners. For outright homeowners, who are mostly older than renters and mortgagors, any changes in the interest rate will have a very small effect, given that the marginal propensity to consume for these households is already relatively low. Interestingly, in the model, renters decrease their consumption following a cut in mortgage and interest rates. We show that through the lens of the model, when solving the dynamic problem, the utility of future homeownership dominates that of a simple consumption-saving motive and prevents renters from dis-saving while facing lower interest rates. Similarly, we also analyze the behavior along the age dimension. In the model, we can replicate empirically documented increase in consumption for young and middle-aged households and very low response for older ones. The access to cheaper mortgages via refinancing choice and lower mortgage payments increases the current consumption of young and middle-aged households. Again, these results align with recent evidence of Wong (2021), which finds that young and middle-aged households have a high concentration of new and refinanced loans.

As documented by Fieldhouse et al. (2018), expansionary credit policy affects not only housing credit markets (such as mortgage rates) but also affects short-term and long-term interest rates (what authors call an “accommodative” monetary policy). Given a rather small number of mortgage market interventions in the data, it is hard to separate the effects on mortgage and interest rates empirically. Our structural model is the perfect environment for that. As such, we analyze an alternative credit policy change that affects only mortgage borrowing rates while the short-term interest rates remain constant. While we find that consumption response in the case of constant interest rates is lower, we still document a significant response of the mortgagors (as well as middle-aged households in general), especially among those who decide to refinance.

Related literature In exploring the link between exogenous credit policy changes and individual consumption, our paper adds to both empirical and theoretical literature on housing and mortgage markets. From the empirical side, we relate to four strands of literature. Firstly, we analyze the US federal government interventions in the mortgage markets. Most of the literature focused on governments' intervention in terms of tax policies. Recent studies include Chambers et al. (2009); Hilber and Turner (2014); Floetotto et al. (2016); Sommer and Sullivan (2018), among others. Fieldhouse et al. (2018) is the most recent study that instead analyzes the interventions to the federal housing agencies rather than any tax policies. In this paper, we use exogenously identified policy interventions from Fieldhouse et al. (2018); unlike Fieldhouse et al. (2018), however, we analyze the transmission mechanisms through which the policy operates using the US household survey data.

Secondly, this paper is related to literature that analyzes the interaction between federal housing agencies and other markets. The most recent studies include Gonzalez-Rivera (2001); Naranjo and Toevs (2002); Lehnert et al. (2008); Hancock and Passmore (2011); Hancock and Wayne Passmore (2014) as well as Fieldhouse et al. (2018). We focus specifically on the effect of mortgage purchases of governmental housing agencies on the consumption of different types of households using a novel identification strategy.

Thirdly, our paper is related to the literature on the role of household balance sheet channels in the transmission of monetary and fiscal policy shocks. These include Iacoviello (2005); Eggertsson and Krugman (2012); Luetticke (2021); Greenwald (2018); Hedlund et al. (2016); Cloyne et al. (2020); Kaplan et al. (2018); Auclert (2019); Bilbiie (2020); Wong (2021), to name a few. Coibion et al. (2017) also uses US household-level data to study the effect of conventional monetary policy on income and consumption inequality. Like in Cloyne et al. (2020), we use the households' housing tenure status to proxy their asset and debt position.

From the theoretical side, our model resembles the recent literature that extends Huggett (1996) model to incorporate housing decisions and aggregate housing and mortgage markets. To name a few, we build on the models of Kaplan et al. (2020); Favilukis et al. (2017); Sommer and Sullivan (2018); Wong (2021), that analyze heterogeneous agents life-cycle economies with uninsurable income risk in which households make a housing and mortgage choice. Unlike these papers, however, we do not focus on the aggregate implications of different macroeconomic shocks but rather analyze the individual households' behavior.

Structure of the paper The rest of the paper is structured as follows. Section 2 sets out the empirical model and presents the impulse response analysis. Section 3 develops a life-cycle economy with endogenous housing choice and uninsurable idiosyncratic risk. Section 4 estimates the model and describes the properties of the baseline economy. Section 5 analyzes the effect of mortgage market intervention within the model framework and discusses transmission mechanisms. Finally, section 6 concludes.

2. Empirical framework

2.1. Institutional background and identification of exogenous policy changes

The total mortgage debt in the US accounts for about 80% of total household debt. For instance, by the 3rd quarter of 2017, the total mortgage debt was about \$8.7 trillion, while auto, credit card, and student debt combined were about \$2.3 trillion. Furthermore, this debt funds the largest asset in the households' net worth - housing.[3]

The US mortgage market is also unique. The US federal government is heavily involved in the mortgage market (especially in terms of residential mortgage purchases) through various agencies: Government-Sponsored Enterprises (GSEs) and Government Agencies (see Fieldhouse and Mertens (2017); Fieldhouse et al. (2018) for detailed history and overview of how these agencies operate). Given the data availability, in this paper, we focus on the involvement of the government through the two largest GSEs: Fannie Mae, founded in 1938 and publicly traded since 1968, and Freddie Mac, founded in 1970. Congress established these agencies to support secondary mortgage markets by buying and securitizing loans from primary mortgage lenders; they are not, however, allowed any direct lending. The share of mortgage debt held by these agencies grew substantially since their establishment and reached a peak of almost 20% by 2004, slowly declining ever since. Fig. 1 plots the agency mortgage holdings as a share of total mortgage debt in the US over time. See Online Appendix for a detailed description of the data.

In the empirical section of this paper, we focus on the portfolio purchases of the housing agencies, shown in the solid blue line in Fig. 2, and how it affects the expenditure of households with different debt position. Unfortunately, simply correlating measures of agency activity with households' expenditure ignores potential endogeneity problems (see Fieldhouse et al. (2018)). On the one hand, Fannie Mae and Freddie Mac respond to market conditions and thus act pro-cyclically. On the other hand, Fannie Mae and Freddie Mac's role is to provide stability in the mortgage markets and thus act counter-cyclically. Ignoring these potential problems makes the causal inference invalid.

To account for the endogeneity in agency market activity, we use a narrative identification approach and major regulatory policy events as an instrument for agency purchase activity. Fieldhouse and Mertens (2017) document significant policy changes that are expected to affect agency portfolios and isolate those events (which they call non-cyclical events) that are free of confounding influences in the spirit of Romer and Romer (2004) and Ramey (2011). These policy changes are

[3] For example, in the 2001 wave of the Survey of Consumer Finances that we use in this paper, the average household held more than 60% of their net worth in housing. Source: Author's calculations.

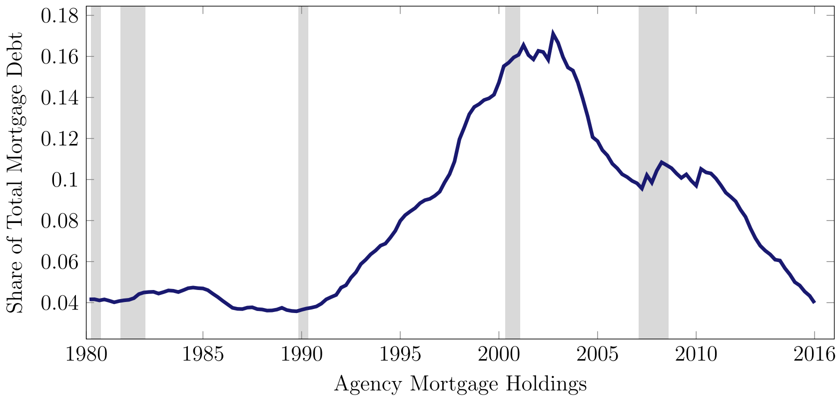

Fig. 1. Agency mortgage holdings as a percent of total mortgage originations. Data is between 1980 and 2016. Grey areas represent NBER recessions. Source: Authors own calculations.

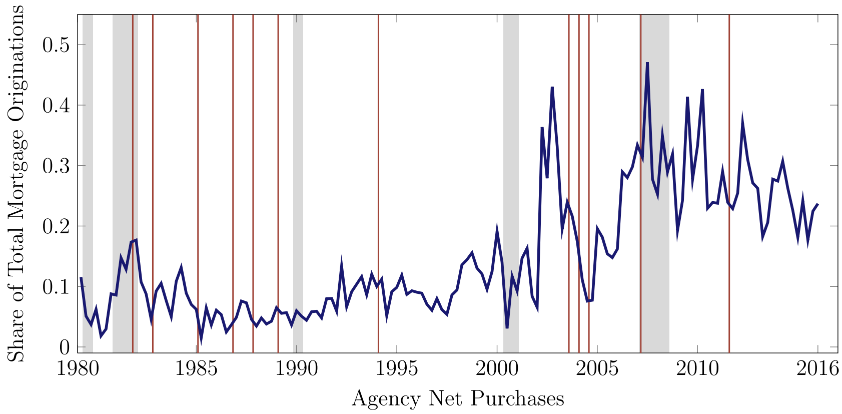

Fig. 2. FNMA & FHLMC net purchase for portfolio investment. Data is between 1980 and 2016. Grey areas represent NBER recessions. Source: Author's own calculations (agency net purchases) and Fieldhouse and Mertens (2017) (policy changes). (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

indicated by vertical red lines in Fig. 2. To quantify these changes, we take a calculated impact of these policy changes from Fieldhouse and Mertens (2017). As in Fieldhouse and Mertens (2017), we normalize this impact value and divide it by the average annualized level of originations in the preceding year. As most of the policy interventions after 2008 were related to the 2007/2008 financial crisis and mainly were cyclically motivated, we limit the analysis to the pre-crisis sample.

2.2. Impulse response specification

To evaluate the effect of agency purchase activity on households' consumption, we conduct an impulse response analysis of shock to agency mortgage purchase. We use local projections instrumental variable approach in the spirit of Jordà (2005) where we use the narrative instrument for identification.

Following Fieldhouse et al. (2018), we start by assessing whether the narrative policy changes do lead to significant changes in net agency purchases. Our first-stage regression specification is of the form

$$\frac{\sum_{j=0}^{h} p_{t+j}}{X_t} = \tilde{a}_h + \tilde{c}_h \frac{\tilde{m}_t}{X_t} + \tilde{d}_h(L) Z_{t-1} + \tilde{u}_{t+h}, \tag{1}$$

where p_t is the agency's net purchase, X_t trend in real mortgage originations, $\tilde{m}_t$ is non-cyclically motivated narrative measure in real dollars, and Z_t is a set of controls (defined below). $\tilde{d}_h(L)$ denotes the polynomial of order 4. Specifically, we focus on the agency purchase activity 2 quarters after the shock, as the robust F-statistic is the highest and equal to 15 (see Online Appendix for details).

We then proceed to identifying the effect of agency purchase activity on variable of interest. Our goal is to estimate the impulse response of the variable of interest y_t (e.g. non-durable expenditure) to increase in the agency purchases. Thus, for a given outcome variable y_t, we estimate the response at horizon k using

$$\log y_{t+k} - \log y_{t-1} = a_k + b_k \left(\frac{4}{2} \times \frac{\sum_{j=0}^{2} p_{t+j}}{X_t} \right) + d_k(L) Z_{t-1} + u_{t+k}, \tag{2}$$

where the term in parentheses denotes net agency purchases made over a 2 quarter period, divided by annualized originations X_t; where the choice of 2 quarters horizon is based on the first-step regression (see above and Online Appendix). We define y_t explicitly below.

The regression in (2) estimates the quarter $k \geq 0$ response to a time 0 news shock to agency purchases. Expected agency purchases are proxied by agency net purchases made over the next half a year ($h = 2$ in (1)). As in Fieldhouse et al. (2018), to address endogeneity, we use the indicator of non-cyclical policy events, deflated by the core PCE price index and scaled by trend originations X_t, as the instrument.

The interpretation of the coefficients b_k is the following. It measures the percentage response of variable of interest y_t to a one percentage point increase in the agencies flow market share (that becomes anticipated 2 quarters ahead).[4]

To make the results comparable to Fieldhouse et al. (2018), we use the same set of control variables Z_t (converted to quarterly values). These are the lagged growth rates of the core PCE price index, a nominal house price index, and total mortgage debt, the log level of real mortgage originations, housing starts, and lags of several interest rate variables: the 3-month T-bill rate, the 10-year Treasury rate, the conventional mortgage interest rate, and the BAA-AAA corporate bond spread. They also include lags of agency net purchases and commitments as a ratio of X_t as well as the unemployment rate and the growth rate of real personal income. See Online Appendix for a detailed description of the data sources and definitions.

2.3. Measuring expenditure data

We use households' expenditure on non-durable goods and services as a response variable y_t in equation (2). To construct our expenditure measure, we use the Consumer Expenditure Survey (CEX) interview section between 1980 and 2007.[5] We define *non-durable goods and services* as food, alcohol, tobacco, fuel, light and power, clothing and footwear, personal goods and services, fares, leisure services, household services, non-durable household goods, motoring expenditure, and leisure goods.[6] We adjust the food at home between 1982 and 1987 following Aguiar and Bils (2015). We also define households' income as income before tax in the past 12 months. After 2005, BLS started imputing missing income observations. Before 2004 we impute missing income observations as in Coibion et al. (2017). We exclude households that are in either top 1% or bottom 1% of either the non-durable expenditure or income distribution. We also exclude households that report zero food expenditure. Finally, we exclude households whose household head is below 25 and over 74 years old. We only keep the households that do not change the housing tenure status between the interviews.

2.4. The effect of agency purchases on expenditure

In this section, we document the response of households' expenditure to shock to agency purchases. The timing is such that the shock can be treated as a news shock to net purchases made over the next half a year.

2.4.1. Response of aggregate expenditure

We first analyze the response of expenditure for all the households using rich expenditure data contained in the CEX survey. Fig. 3 plots the percentage response of aggregate consumption expenditure to a one percentage point increase in the agencies flow market share anticipated two quarters before. One year after the shock, aggregate consumption increased by about 0.04 percent, and the increase is significant at 95% confidence level. The response remains significant for up to 8 quarters after the shock, slowly declining after two years.[7]

2.4.2. Pseudo-cohort analysis by housing tenure status

As documented by Fieldhouse et al. (2018), an increase in agency mortgage purchases boosts mortgage lending and lowers mortgage rates. It is, therefore, important to distinguish between those households who own the house with a mortgage and those without. In the long run, agency purchases also influence house prices and expand homeownership. Therefore the effect on those households who own the house and those who do not might be different. The CEX survey,

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[4] The average market share of the agencies was about 7% between 1967 and 1990 and increased to about 15% by the end of 2006, see Fieldhouse et al. (2018).

[5] Data between 1980 and 1995 is obtained from ICPSR through UK Data Service. Post-1995 data is publicly available at the Bureau of Labor Statistics (BLS) website.

[6] We omit the durable consumption (that accounts for around 17% of total expenditure, a number consistent with that reported in Aguiar and Bils (2015)) from the analysis for several reasons. First and foremost, the mapping between consumption and expenditure for durable goods is no longer direct: durable goods provide consumption for more than one period. Secondly, introducing durable consumption into our structural model in Section 3 would increase the dimensionality of the model and would complicate the analysis. Yet, in Online Appendix C, we repeat all the empirical analyses for durable consumption.

[7] Fieldhouse et al. (2018) find a modest increase in consumption a year after the shock. While we find larger magnitude of the response using micro data on consumption in the CEX, in terms of timing of the response - our results are consistent with those of Fieldhouse et al. (2018) as they show a delayed response in the consumption of around one year.

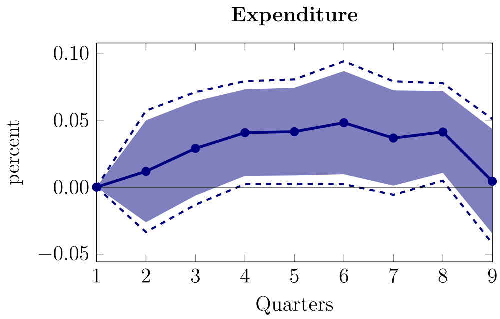

Fig. 3. Impulse response of total non-durable expenditure a 1% increase in net purchase by FNMA & FHLMC, anticipated 2 quarters before. Blue areas and broken lines represent 90% and 95% confidence intervals, respectively.

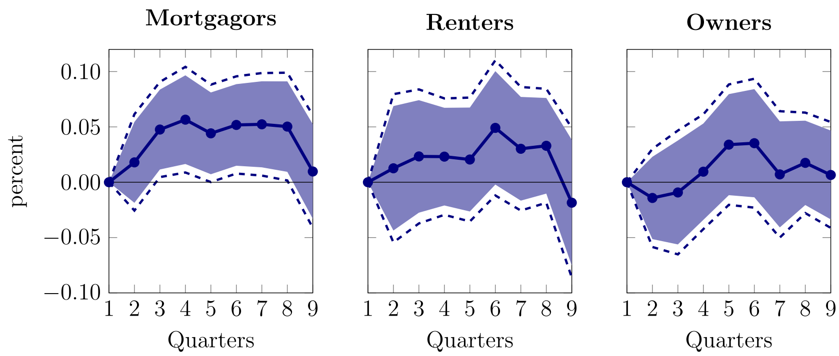

Fig. 4. Impulse response of non-durable expenditure by housing tenure status a 1% increase in net purchase by FNMA & FHLMC, anticipated 2 quarters before. Blue areas and broken lines represent 90% and 95% confidence intervals, respectively.

on top of containing rich expenditure data, contains information on housing tenure status. We utilize this information and group the households into three categories based on their tenure status in the spirit of Cloyne et al. (2020). The categories are *renters*, *mortgagors*, and *outright owners*. Unfortunately, given the rotating panel nature of the CEX survey, it is impossible to follow individual households for more than four quarters over which they are observed. We, therefore, employ a grouping estimator to aggregate individual observations into pseudo-cohorts by housing tenure as in Browning et al. (1985). One concern with this identification strategy is the endogenous switching between housing tenure status because of policy changes. In the CEX, we can track exactly the households that switch between the interviews. We find that a very small number of households actually do switch the status between the interviews. Moreover, we find that increased agency purchases do not lead to endogenous switching of tenure status in the CEX.[8]

As such, we look at the response of households' expenditure, based on their housing tenure status, to a one percentage point increase in the agencies flow market share, anticipated two quarters in advance, under the specification in (2) using the non-cyclically motivated narrative measure as an instrument. Fig. 4 plots the results for three different groups of households. The left panel plots the impulse response (together with 90% and 95% confidence intervals) of mortgagors, the middle panel that of the renters, and the right panel that of the outright homeowners. From the figure, we see that response of mortgagors is large and significant between three and eight quarters after the shock, reaching a peak of about 0.06 percent four quarters after, slowly declining after two years. Renters also increase their expenditure (reaching a value of around 0.05 percent six quarters after the shock), but the increase is not statistically significant at all horizons. Finally, outright homeowners have the smallest increase in non-durable expenditure (in fact, they initially exhibit a slight decrease in expenditure), and the increase is insignificant for all horizons up to two years.

2.4.3. Analysis by age

Homeownership and mortgage status exhibit a strong life-cycle pattern: most renters are young, mortgagors are primarily young and middle-aged, while outright homeowners tend to be older. To understand to what extent the results in Fig. 4 are driven by the age dimension, instead of looking at the housing tenure status, we divide the households into three age categories: young (those aged between 25 and 34), middle-aged (those aged between 35 and 64), and old (those aged

[8] In the Online Appendix we report some of the statistics on those households, as well as analyze if the increase in agency purchases leads to endogenous switching between the housing tenure status.

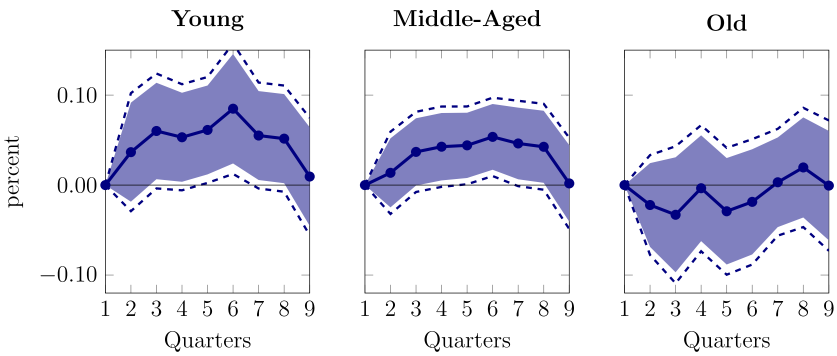

Fig. 5. Impulse response of non-durable expenditure by age a 1% increase in net purchase by FNMA & FHLMC, anticipated 2 quarters before. Blue areas and broken lines represent 90% and 95% confidence intervals, respectively.

between 65 and 75).[9] To be consistent with the results above, the hypothesis would be that young, and middle-aged households (those who are most likely to be mortgagors, as well as renters) would increase their consumption the most. Indeed, the results reported in Fig. 5 confirm this hypothesis. Young households (left panel in Fig. 5) increase their consumption significantly, reaching a peak of almost 0.08 percent six quarters after the shock. Similarly, middle-aged households also increase their consumption, which is significant between four and eight quarters after the shock. These responses are consistent with the increase in expenditure by mortgagors and renters (even though the latter is not significant). Finally, older households (right panel) do not respond significantly and slightly decrease their expenditure in the initial quarters.

Online Appendix plots impulse response of non-durable expenditure interacting both the age and the housing tenure status, and extends the analysis to durable expenditures.

2.4.4. Comparing the results to monetary policy shocks

As documented by Fieldhouse et al. (2018), an increase in the agencies' flow market share (through an increase in mortgage purchases by the agencies) lowers mortgage rates (by about 10-15 basis points) as well as short-term and long-term lending rates (by about ten basis points). To understand the magnitude of the responses of consumption we documented above, as well as the timing of the response of the households, we can partially compare the results to the existing literature that analyzes heterogeneous responses of consumption expenditure (see Cloyne et al., 2020) to exogenous monetary policy shocks. Cloyne et al. (2020) find that the mean estimate for mortgagors of about 0.1-0.2 percent for a 25bp cut in the policy rate, the response that becomes significant around nine quarters after the initial shock. In terms of magnitudes, for an equivalent cut in the short-term lending rates, our results would translate to a response of mortgagors of around 0.1-0.15 percent, the numbers comparable and similar to those by Cloyne et al. (2020). Regarding timing, however, we find the significant response of mortgagors already after four quarters after the initial shock. One of the explanations of why the response in expenditure for mortgagors to increase in agency purchases is speedier is the refinancing channel that we carefully analyze in the context of the structural model later on. Indeed, this channel is also consistent with Fieldhouse et al. (2018) (see Figure V), which finds a large and significant increase in mortgage refinancing shortly after the increase in agency purchases. Regarding the responses of other categories of households, the dynamic response of the other two housing tenure categories is consistent with that of Cloyne et al. (2020). We also find the insignificant response of outright homeowners, with an initial decrease in expenditure, pointing to the idea that outright homeowners might no longer be sensitive to the changes in the mortgage rates (as well as the interest rates). With regards to the response of renters, as they should not be responding to changes in the mortgage rates and rather interest rates only, the dynamic response of renters following an increase in agency purchases (and how it translates to borrowing and saving rates) is consistent to that of a response to monetary policy shocks.

3. A life-cycle model with housing markets

In the previous section, we documented the causal effect of news shock to agency mortgage purchases. Unfortunately, with the data available, it is impossible to identify the transmission mechanism through which the effects work. To understand which channel is exactly responsible for an increase in expenditure for mortgagors only and an increase in the inequality for that group, in this section, we develop a Huggett (1996) type of heterogeneous agent life-cycle model with uninsurable risk, endogenous housing choice, and aggregate mortgage market shocks (see Sommer and Sullivan (2018); Kaplan et al. (2020); Wong (2021)). Through the lens of this model, we explain the empirical evidence we found in the previous sections and analyze which channels contribute to the results indicated.

[9] We are no longer concerned about endogenous switching between the different groups of households, as a change in the age of the household is entirely orthogonal to our series of shocks.

3.1. Demographics, preferences and labor income

Demographics Time is discrete. The economy is populated by a continuum of finitely-lived households. Age is indexed by $j = 1, \ldots, J$. Households work for first $J_r - 1$ periods and are retired until period J. Every period households face the risk of early death. Let s_j denote the probability of survival to age j conditional to surviving to age $j - 1$. All households die with certainty at age J.[10] The demographics in the model are stable: there is a constant population growth n, and all age-j agents make a constant fraction of population μ_j.[11]

Preferences Households maximize expected lifetime utility is given by

$$\mathbb{E}_0 \left[\sum_{j=1}^{J} \beta^{j-1} \left(\prod_{k=1}^{j} s_k \right) u_j(c_j, s_j) + \beta^J v(a_J) \right] \tag{3}$$

Here c_j denotes the consumption of goods at age j and s_j denotes the consumption of housing services at age j, β is the discount factor, and a_J is the amount of bequest left by a household of age J. The only source of uncertainty in the economy is the idiosyncratic income shock (described below). Utility function u is an aggregator over consumption and housing

$$u_j(c_j, s_j) = e_j \frac{\left[c_j^{\alpha} s_j^{1-\alpha} \right]^{1-\vartheta} - 1}{1 - \vartheta} \tag{4}$$

while the bequest function v is given as in De Nardi (2004)

$$v(a) = \psi \frac{(a)^{1-\vartheta} - 1}{1 - \vartheta}, \tag{5}$$

where α denotes housing preference, ϑ is the risk aversion, and ψ measures the strength of bequest motive. Here, e_j is the exogenous equivalence scale that captures exogenous life-cycle changes in household structure and composition.

Labor income Working-age households receive exogenous income y_j given by

$$\log y_j = \chi_j + \epsilon_j, \tag{6}$$

where χ_j is the deterministic age profile and ϵ_j is the idiosyncratic component that follows first-order Markov process. After retirement, households receive social security benefits

$$y_j = \rho_{ss} y_{J_r}, \quad j > J_r,$$

where ρ_{ss} is a replacement rate and y_{J_r} are their earnings in the last working period.[12] The pay-as-you-go social security system is run by the government.

3.2. Housing

Households can either rent or own a house. Houses are characterized by their size, which is given by a discrete set. Let $\widetilde{\mathcal{H}}$ denote the set of houses available for rent, while $\mathcal{H}$ denotes the set of owner-occupied houses. We assume that the price of the house is fixed and is equal to p_h while the rental price of a housing unit is denoted by ρ_h, which is equal to a fixed fraction of the house price.

To distinguish house owners from house renters, we assume that housing generates service flow equal to the size of the house for renters, i.e., $s = \tilde{h}$, where $\tilde{h} \in \widetilde{\mathcal{H}}$, while owning a house generates an extra utility for the household, such that $s = \omega h$, where $\omega > 1$ and $h \in \mathcal{H}$.

Every period the homeowner has to pay the maintenance cost $\delta_h p_h h$ that fully offsets the physical depreciation of the house and property taxes $\tau_h p_h h$. There is a transaction cost equal to $\kappa_h p_h h$ associated with buying or selling a house. There is no transaction cost for renting.

3.3. Assets, mortgages and market arrangements

Liquid assets Households can save in one-period bonds, a, with an exogenous interest rate given by r_a. Households are not allowed any unsecured borrowing, and their borrowing constraint is given by $a \geq 0$.

Let q_a denote the price of bond, such that $q_a = 1/(r_a + 1)$

[10] This implies that $s_{J+1} = 0$. Also, $s_1 = 1$.

[11] This measure is defined as $\mu_{j+1} = s_{j+1} \mu_j (1 + n)$, with μ_1 such that $\sum_{j=1}^{J} \mu_j = 1$, so that the total population is fixed to 1.

[12] To reduce the numerical burden, we only keep track of the last period's income to determine the household's pension. Alternatively, at the cost of increasing the state space in the model, we can include the whole history of income.

Mortgages House purchases can be financed by a mortgage. A household that takes out a new mortgage with a principal balance m'. We assume that all mortgages are long-term, subject to interest rate r_m, and have to be repaid over the remaining working life of the borrower. We assume that mortgage rate r_m is given by

$$r_m = (1+\iota)r_a, \tag{7}$$

where ι controls the spread between r_a and r_m. Down-payment for a borrower who takes out a mortgage of size m' to buy a house of size h' is

$$p_h h' - m' \tag{8}$$

Mortgage origination is also subject to a fixed origination cost κ_m. When taking out a mortgage, households have to satisfy two constraints. The first one is the maximum loan-to-value constraint: the initial mortgage size must be less than a fraction λ_m of the value of the house being purchased

$$m' \le \lambda_m p_h h' \tag{9}$$

The second constraint is the maximum payment-to-income constraint: the first minimum mortgage payment must be less than a fraction λ_π of the income at the time of purchase

$$\pi_j^{min}(m') \le \lambda_\pi y_j, \tag{10}$$

where we define the minimum payment function $\pi_j^{min}(m')$ using a constant amortization formula

$$\pi_j^{min}(m') = \frac{r_m(1+r_m)^{J-j}}{(1+r_m)^{J-j}-1} m' \tag{11}$$

that assumes that the borrower is required to make $J-j$ payments π that exceed the minimum payment requirement after mortgage origination. The remaining mortgage principle evolves according to

$$m' = m(1+r_m) - \pi \tag{12}$$

We also assume that households are allowed to refinance the existing mortgage. When refinancing (taking out a new mortgage), households have to repay the existing mortgage balance, pay the fixed mortgage origination cost, and satisfy both loan-to-value and payment-to-income constraints. Households are also allowed to sell the house, given that they repay the remained of the mortgage as well as transaction costs.

3.4. Government

In the model, the government receives revenues from the property tax τ_h and progressive income tax $\mathcal{T}(y,m)$ that depends on income y and mortgage holdings m. Interest paid on mortgages is deductible up to a predetermined threshold. We assume that tax function is progressive as in Heathcote et al. (2017) and $\mathcal{T}$ takes the form

$$\mathcal{T}(y,m) = y - \tau_y^0 \left(y - r_m \min\{m, \bar{m}\}\right)^{1-\tau_y^1} \tag{13}$$

where τ_y^0 and τ_y^1 measure the progressivity of the tax system and $\bar{m}$ denotes the maximum allowed deductible mortgage. On the spending side, the government finances the social security system for the households. The government runs a balanced budget, with services G (not valued by the household) adjusting to absorb any difference between government income and spending.

3.5. Dynamic programming problem of the household

We now describe the dynamic programming problem of households. There are two types of households in the economy: homeowners and non-homeowners. Let V_j^n denote the value function of the non-homeowner at age j and let V_j^h denote the value function of the homeowner at age j. When non-homeowner enters the period at age j he has two choices - either remain a non-homeowner in the next period (rent a house) or become a homeowner next period (buy a house). Let V_j^r and V_j^o denote the value function of renters and buyers, respectively. Non-homeowners essentially solve the following problem

$$V_j^n(\mathbf{x}_j^n) = \max\left\{V_j^r(\mathbf{x}_j^n), V_j^o(\mathbf{x}_j^n)\right\} \tag{14}$$

where $\mathbf{x}_j^n$ denotes the vector of state variables of the non-homeowner, described below. Above, V_j^o denotes the value function of a renter who chooses to become a homeowner (hence his state variables are $\mathbf{x}_j^n$), while V_j^h is a value function of the current homeowner (and so he enters the period with a house and as such is different from a new buyer, please see below).

When homeowner enters the period, he has three different choices. He can either continue paying the existing mortgage (let V_j^p denote the value function of the mortgage payer), repay the existing mortgage and get a new mortgage (let V_j^f denote the value function of the mortgage refinancer), or repay the remaining mortgage and sell the house (let V_j^s denote the value function of the seller). Every period, the homeowner solves the following problem

$$V_j^h(\mathbf{x}_h) = \max\left\{V_j^p(\mathbf{x}_j^h), V_j^f(\mathbf{x}_j^h), V_j^s(\mathbf{x}_j^h)\right\} \tag{15}$$

where $\mathbf{x}_j^h$ denotes the vector of state variables of the homeowner, described below.

Non-homeowners of age j enter the period with a holding of liquid assets a_j and exogenous income y_j. Homeowners of age j, on the other hand, also enter the period with an outstanding balance on the mortgage m_j and house h_j. When $m_j > 0$ we refer to homeowners as the mortgagor, whereas when $m_j = 0$ we refer to them as outright owners. Thus

$$\mathbf{x}_j^n = \left(a_j, y_j\right) \tag{16}$$

$$\mathbf{x}_j^h = \left(a_j, m_j, h_j, y_j\right) \tag{17}$$

We now describe in detail the problem of each household in a recursive form. From here on, the state and control variables with no subscript denote the current age/period variables, i.e. $a_j = a$, while state and control variables with $'$ superscript denote the next period/age variables, i.e. $a_{j+1} = a'$.

Renters The households of age j that enter the period as non-homeowners and decide to rent next period choose the level of consumption today (c), the level of liquid savings next period (a') and the size of the rented dwelling for the next period ($\tilde{h}'$). In recursive form, their problem can be written as

$$V^r(\mathbf{x}^n) = \max_{c,a',\tilde{h}'} u(c,s) + \beta\mathbb{E}_{\epsilon'}\left[V^{n'}\left(\mathbf{x}^{n'}\right)\right] \tag{18}$$

Renters solve the above problem subject to:

$$\begin{aligned}
&c + \rho_h\tilde{h}' + q_a a' \le a + y - T(y,0) \qquad (19)\\
&a' \ge 0\\
&s = \tilde{h}', \quad \tilde{h}' \in \widetilde{\mathcal{H}}\\
&y' \sim Y(y)
\end{aligned}$$

where the equations above are a budget constraint, borrowing constraint, housing services production, and income evolution, respectively. Let $\mathbb{1}^r\left(\mathbf{x}^n\right)$ denote the decision of a non-homeowner with state variables $\mathbf{x}^n$ to rent a house.

Buyers The households of age j that enter the period as non-homeowners and decide to buy a house choose the level of consumption today (c), the level of liquid savings next period (a'), the size of the house to buy (h'), and the level of mortgage to take out. In recursive form, their problem can be written as

$$V^o(\mathbf{x}^n) = \max_{c,a',h',m'} u(c,s) + \beta\mathbb{E}_{\epsilon'}\left[V^{h'}\left(\mathbf{x}^{h'}\right)\right] \tag{20}$$

Renters solve the above problem subject to:

$$\begin{aligned}
&c + q_a a' + p_h h' + \kappa_m \le a + y - T(y,0) + m' \qquad (21)\\
&m' \le \lambda_m p_h h'\\
&\pi^{\min}(m') \le \lambda_\pi y\\
&a' \ge 0\\
&s = \omega h', \quad h' \in \mathcal{H}\\
&y' \sim Y(y)
\end{aligned}$$

where the new equations are the LTV and PTI constraints. Let $\mathbb{1}^o\left(\mathbf{x}^n\right)$ denote the decision of non-homeowner with state variables $\mathbf{x}^n$ to buy a house, with

$$\mathbb{1}^r\left(\mathbf{x}^n\right) + \mathbb{1}^o\left(\mathbf{x}^n\right) = 1$$

Mortgage payers The households of age j that enter the period as homeowners with a given level of mortgage m and house size h, and decide to make the payment towards the mortgage balance, choose the level of consumption today (c), the level of liquid savings next period (a'), and the size of payment (π). In recursive form, their problem can be written as

$$V^p(\mathbf{x}^h) = \max_{c,a',\pi} u(c,s) + \beta \mathbb{E}_{\epsilon'} \left[V^{h'} \left(\mathbf{x}^{h'} \right) \right] \tag{22}$$

Mortgage payers solve the above problem subject to:

$$\begin{aligned}
& c + q_a a' + (\delta_h + \tau_h) p_h h' + \pi \leq a + y - T(y,m) \\
& m' = (1 + r_m) m - \pi \\
& \pi \geq \pi^{\min}(m) \\
& a' \geq 0 \\
& s = \omega h', \quad h' = h \in \mathcal{H} \\
& y' \sim Y(y)
\end{aligned} \tag{23}$$

where the new equations are the mortgage balance evolution and minimum payment requirement. When choosing the current level of mortgage payment, the household needs to satisfy the minimum payment requirement. This constraint is similar to the PTI requirement on origination (equation (10)), but does not limit household making payments out of the income only (for example, a household that received a negative income shock can still make a mortgage payment by using some of the assets). Let $\mathbb{1}^p \left(\mathbf{x}^h \right)$ denote the decision of the homeowner with state variables $\mathbf{x}^p$ to make a payment towards the mortgage.

Mortgage refinancers The households of age j that enter the period as homeowners with a given level of mortgage m and house size h, and decide to refinance the existing mortgage, choose the level of consumption today (c), the level of liquid savings next period (a'), and the level of new mortgage (m'). In recursive form, their problem can be written as

$$V^f(\mathbf{x}^h) = \max_{c,a',m'} u(c,s) + \beta \mathbb{E}_{\epsilon'} \left[V^{h'} \left(\mathbf{x}^{h'} \right) \right] \tag{24}$$

Mortgage refinancers solve the above problem subject to:

$$\begin{aligned}
& c + q_a a' + (\delta_h + \tau_h) p_h h' + (1 + r_m) m + \kappa_m \leq a + y - T(y,m) + m' \\
& m' \leq \lambda_m p_h h' \\
& \pi^{\min}(m') \leq \lambda_\pi y \\
& a' \geq 0 \\
& s = \omega h', \quad h' = h \in \mathcal{H} \\
& y' \sim Y(y)
\end{aligned} \tag{25}$$

Let $\mathbb{1}^f \left(\mathbf{x}^h \right)$ denote the decision of the homeowner with state variables $\mathbf{x}^p$ to refinance the existing mortgage.

Sellers The households of age j that enter the period as homeowners with a given level of mortgage m and house size h, and decide to sell their house in the current period, choose the level of consumption today (c), the level of liquid savings next period (a') and the size of the rented dwelling for the next period ($\tilde{h}'$), as they will remain non-homeowners for the following period.

$$V^s(\mathbf{x}^h) = \max_{c,a',\tilde{h}'} u(c,s) + \beta \mathbb{E}_{\epsilon'} \left[V^{n'} \left(\mathbf{x}^{n'} \right) \right] \tag{26}$$

House sellers solve the above problem subject to:

$$\begin{aligned}
& c + \rho_h \tilde{h}' + q_a a' \leq a_s + y - T(y,m) \\
& a' \geq 0 \\
& s = \tilde{h}', \quad \tilde{h}' \in \widetilde{\mathcal{H}} \\
& y' \sim Y(y)
\end{aligned} \tag{27}$$

where a_s denotes the current level of assets plus the proceedings from selling the house net of transaction costs and mortgage balance, given by

Table 1
Targeted and non-targeted moments in the estimation.

Targeted Moments		
Moment	Model Value	Empirical Value
Homeownership rate	0.68	0.67
Fraction of homeowners with gross mortgage debt	0.61	0.67
Median house value-to-income ratio	3.80	3.61
Non-Targeted Moments		
Moment	Model Value	Empirical Value
Average earnings owners to renters	2.15	2.32
Wealth-to-income ratio	3.44	3.54
Homeownership rate at 30yo	0.47	0.47
Median Loan-To-Value	0.63	0.59
Median Loan-To-Income	2.03	1.94

$$a_s = a + (1 - \delta_h - \tau_h - \kappa_h) p_h h - (1 + r_m) m. \tag{28}$$

Let $\mathbb{1}^s\left(\mathbf{x}^h\right)$ denote the decision of the homeowner with state variables $\mathbf{x}^h$ to sell the house.

Recall that for the homeowner with state variables $\mathbf{x}^h$

$$\mathbb{1}^p\left(\mathbf{x}^h\right) + \mathbb{1}^f\left(\mathbf{x}^h\right) + \mathbb{1}^s\left(\mathbf{x}^h\right) = 1$$

Online Appendix D provides the formal definition of the equilibrium of the numerical model.

4. Parametrization

The model is parametrized at an annual frequency. There are three sets of parameters in the model. A subset of parameters is set exogenously without the need to solve for the steady-state of the model. These include parameters such as the deterministic profile of earnings, tax parameters, survival probabilities, and equivalence scale (see Online Appendix). A set of parameters are taken from the literature and are described below. The remainder of the parameters is estimated using the Simulated Method of Moments to match a set of moments taken from the 2001 wave of SCF. The target model-implied and data moments are reported in Table 1. The model-implied average homeownership rate is 68%, compared to 67% in the data; the model generates around 61% of homeowners with gross mortgage debt, several percentage points below the data value of 67%. Finally, the median house value-to-income ratio is 3.8 in the model, compared to 3.61 in the data. We describe the parameters used in the estimation below. We also report several non-targeted moments. Those are the average earnings of owners to renters (model implied value of 2.15 compared to 2.32 in the data), total wealth-to-income ratio (3.44 vs 3.54), and homeownership rate at the age of 30 (47% both in the model and in the data). Also, the model implied median loan-to-value and loan-to-income values are only slightly above the data values (0.63 vs 0.59 and 2.03 vs 1.94, respectively). Please see Online Appendix for the definitions of the variables used to construct data moments.

Demographics and preferences The model period is one year. Households enter the economy in age 21, retire at age 60 and live until age 81. This corresponds to $J_r = 40$ and $J = 60$.

We set risk aversion parameter ϑ equal to 2.0 so that the EIS is 0.5. The properties of the baseline model are robust to change in ϑ as long as EIS is less than 1.

The strength of the bequest motive is controlled by ψ, which we set equal to 50. In the model, this value is not well identified with the typical targets used in the literature. As such, setting it to a positive number guarantees that households do not sell the house in the last period of life.[13]

Finally, we pick three remaining preference parameters (α, β and ω) to estimate three model-implied moments. In particular, we pick the average homeownership rate, the average share of homeowners with gross mortgage debt, and the median house value-to-income ratio as our targets. The model-implied median house value-to-income ratio is 3.80 compared to 3.61 in the data. The model-implied fraction of homeowners with gross mortgage debt is 61 percent compared to 67 percent in the data. The model-implied homeownership rate is 68 percent compared to 67 percent in the data. The implied estimates are $\beta = 0.93$, $\omega = 1.5$, and $\alpha = 0.67$. All demographic and preference parameters are summarized in Table 2.

Labor income and government expenditure The deterministic and stochastic component of labor earnings, χ_j, is calculated using the data on labor earnings from the 2001 wave of the SCF. The stochastic component of earnings is modeled as an

[13] In particular, we have experimented with including ψ as one of the estimated parameters and including either the ratio of net worth at age 75 to net worth at age 50, aggregate net worth over income or ratio between 75th and 25 percentile of net worth for retired households as one of the moments. In general, none of those moments helped to precisely identify ψ.

Table 2
Parameter values (demographics and preferences).

Demographics and Preferences			
Parameter	Meaning	Exogenous	Value
J	Length of life	Y	60
J_r	Working life	Y	40
ϑ	Risk aversion	Y	2.0
β	Discout factor	N	0.93
ψ	Strength of bequest	Y	50
ω	Utility from homeownership	N	1.5
α	Share of housing in utility	N	0.67
s_j	Survival Probability	Y	See Online Appendix E
e_j	Equivalence Scale	Y	See Online Appendix E

Table 3
Parameter values (labor income and government expenditure).

Labor Income and Government Expenditure			
Parameter	Meaning	Exogenous	Value
χ_j	Deterministic life-cycle profile	Y	Online Appendix E
τ_y^0	Income tax parameter	Y	0.75
τ_y^1	Income tax parameter	Y	0.151
ρ_{ss}	Replacement rate	Y	0.6
$\bar{m}$	Mortgage deduction limit	Y	20*
τ_h	Property tax	Y	0.01

* A unit of the final good corresponds to $50000, which is the median income in the 2001 wave in SCF.

AR(1) process with a persistence of 0.97 and standard deviation of 0.20, and a standard deviation of the initial distribution of income of 0.42 (see Kaplan et al. (2020)). We set the social security replacement rate to 60 percent. The parameters of the tax function (13), τ_y^0 and τ_y^1, are set to 0.75 and 0.151, respectively, and are taken from Heathcote et al. (2017) for the US. Parameter τ_y^0 measures the average level of taxation, and parameter τ_y^1 measures the degree of progressivity. The maximum level of tax-deductible mortgage, $\bar{m}$, is set to correspond to $1 million. The property tax τ_h is set to 1 percent, which is the median tax rate across US states (see, for example, Tax Policy Center). These are summarized in Table 3.

Housing We fix the grid for the owner-occupied houses ($\mathcal{H}$) and rented houses ($\widetilde{\mathcal{H}}$), so that households are only allowed to choose to buy or rent of the dwellings from a grid. The minimum size of the owner-occupied dwelling partially controls the homeownership rate for the young (it acts as a barrier to entry). As such, we set the minimum house size to partially match the homeownership rate for households at age 30. The model-implied value is 47 percent, compared to 47 percent in the data. The depreciation rate of housing is set equal to 1.5 percent. The transaction cost of selling the house, κ_h, is set to 8 percent, which is the average value reported in Quigley (2002). These are summarized in Table 4.

Liquid assets and mortgages We set the interest rate r_a exogenously equal to 2.5 percent, and the spread parameter ι equal to 30 percent. This implies the mortgage rate r_m of about 3.2 percent. The spread is set to match the observed difference between the average rate on 30-year fixed-term mortgages (series MORTGAGE30US from FRED) and the 10-year T-Bill rate (series GS10) in 2001. The implied price of the bond, q_a is equal to 0.975. The mortgage origination cost, κ_m, is set to the equivalent of $ 2000, corresponding to the sum of application, attorney, appraisal, and inspection fees (see Kaplan et al. (2020)).[14] These are summarized in Table 4.

4.1. Model fit

As reported above, to estimate some of the internal parameters in the model (discount rate, share of consumption in the utility as well as preference for owning vs renting), we run the simulated method of moments to target three moments of interest: average homeownership rate, the average share of homeowners with gross mortgage debt, and median house

[14] Parameter κ_m acts as a fixed cost both for new homeowners as well as households who decide to refinance. Given that the model has no aggregate uncertainty, in the model, households only refinance when facing individual income uncertainty. As such, it is not straightforward to map the share of households that refinance the existing mortgage from the data to the model. The closest reference of how many households refinance in a given period would be Boar et al. (2022), which reports a number of 8%. In our estimation, the implied share of households who refinance is around 5%, slightly below this value.

Table 4
Parameter values (housing, liquid assets and mortgages).

Housing, Liquid Assets and Mortgages			
Parameter	Meaning	Exogenous	Value
δ_h	Depreciation rate	Y	0.015
κ_h	Transaction cost	Y	0.08
r_a	Interest rate	Y	0.025
ι	Spread	Y	0.3
r_m	Mortgage rate	Y	0.032
q_a	Price of bond	Y	0.975
κ_m	Mortgage origination cost	Y	0.04[*]
λ_a	Maximum borrowing limit	Y	0
q_m	Down payment requirement	Y	0.15

[*] A unit of the final good corresponds to $50000, which is the median income in the 2001 wave in SCF.

to income ratio. These moments are the estimates of the average behavior of the households in the model. In Fig. 6, to check the overall fit of the model, we plot the implied life-cycle profiles that the model produces. In particular, for both the data (using the 2001 wave of the SCF) and the model, we report the average homeownership rate over life-cycle, the average share of households with the mortgage, median leverage (ratio of debt to house value for the homeowners with the mortgage), median debt to income values, and median housing wealth to income values. We also report the model-generated average income and consumption profiles. Note that none of those life-cycle profiles were directly targeted in the estimation.

The model slightly overestimates the homeownership rate for middle-aged households (between the ages of 35 and 65) - implying that we also overestimate the share of households that take out a mortgage in the model - see the right top panel in Fig. 6). One of the possible explanations could be that the model terminates in period 60 (that corresponds to age 80), and the households in the model are forced to repay any remaining mortgage until the last period; in the data, on the other hand, there are households above the age of 80 that hold a positive value of the mortgage. Regarding the other life-cycle profiles, the model generates a decreasing median leverage (again, since the households are required to deleverage at the end of life, the decrease is steeper post-retirement), and relatively flat profiles of loan-to-income and housing-to-income. Finally, the middle right panel also plots the life-cycle profiles of income and consumption. With regards to income, it is an exogenous input into the model and is assumed to remain flat after retirement; consumption exhibits a hump-shaped profile, partially reflecting the presence of the equivalence scale in order to capture the change in the household structure over the life-cycle.

5. Mortgage market intervention experiment

We next perform a mortgage market intervention experiment in the baseline model using the empirical evidence on the effects of governmental mortgage markets interventions on interest and mortgage rates.

5.1. Macroeconomic effects of mortgage market intervention

In their paper, Fieldhouse et al. (2018) document the macroeconomic effects of news shock to agency mortgage purchases. They find that following a shock, the interest rates, as well as the mortgage rates, decrease, as does the spread between mortgage rates over the interest rates. In particular, they show that mortgage rates decrease by about 10-15 basis points a year after the shock (that maps to 1 period in our model), while the short and long-term lending rates drop by about 10 basis points. They also document a drop in the spread between mortgage and lending rates. As emphasized by Fieldhouse et al. (2018), this simultaneous drop in both rates is an evidence that agency mortgage purchases are accompanied by accommodative monetary policy. We describe how we map this empirical evidence into our model in the next section.

Fig. 7 plots the impulse responses of interest rates and mortgage rates that we feed as the policy variables into the model. Those figures are a smooth version of the impulse responses found in Fieldhouse et al. (2018) for FHA mortgage rate and 10-year treasury rate (see Figure VII in Fieldhouse et al. (2018)), evaluated at the annual frequency (that corresponds to the period of the model).

Fieldhouse et al. (2018) also find that following an increase in agency mortgage purchases, house prices rise gradually but very persistently over time, but this increase in house prices is relatively small and imprecisely estimated. In Online Appendix we repeat the analysis below where we also exogenously change house prices (together with interest rates) as a result of an increase in agency purchases. In what follows below, however, we focus solely on changes in interest rates (both short-term lending rate and mortgage rate).

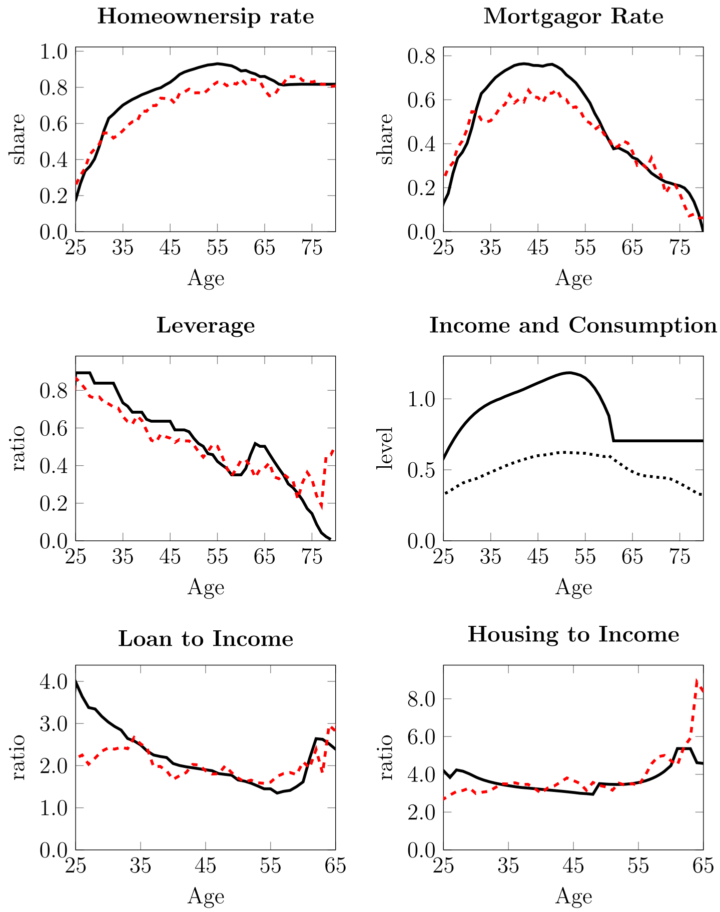

Fig. 6. Mean life-cycle profiles in the baseline model. Top left panel displays mean homeownership rate. Top right panel displays mean share of households with the mortgage. Middle left panel plots the median leverage (defined as debt over housing wealth). Middle right panel plots the model generated income and consumption profiles. Lower left panel plots median loan to income ratio. Lower right panel plots median housing wealth to income ratio. Where applicable, black solid line corresponds to model-generated life-cycle profiles, while red dashed line corresponds to data generated profiles.

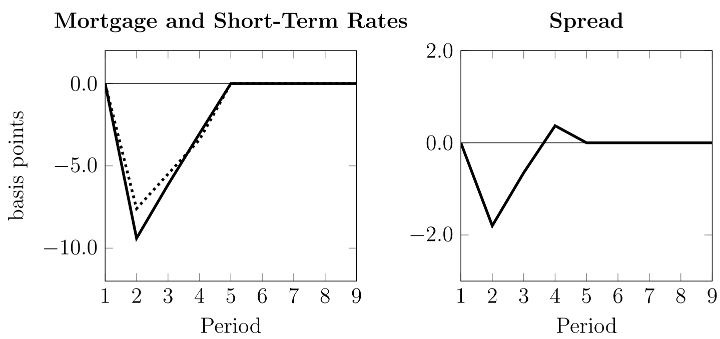

Fig. 7. Left panel plots the path of short-term interest rate (dotted line) and mortgage rate (solid line) that we feed into the structural model. Right panel plots the implied change in spread between the two.

5.2. Transitional dynamics

To understand why households respond with an increase in consumption following an increase in agency purchases, we will use the estimated structural model presented in the previous section and compute transitional dynamics treating the government intervention to the mortgage markets (and an increase in agency purchases that follows) as an unexpected shock to the steady-state in the economy. In particular, we will model two simultaneous transitory socks - to the short-term lending interest rate and to long-term mortgage rate - as shown in Fig. 7 - and compute changes in consumption relative to the initial steady-state of the economy.

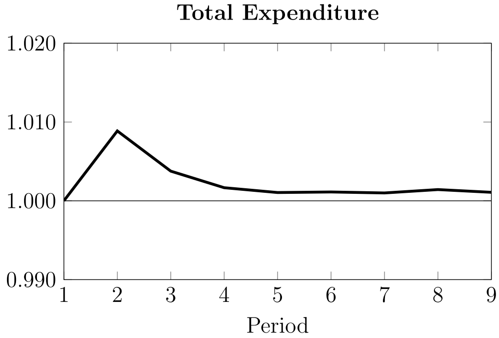

Fig. 8. The figure plots the simulated path of average expenditure for all households. Initial period is normalized to 1.

We then analyze whether the policy experiment can reconcile the empirical evidence presented in section 2.4. Empirically, we found several important results. First, we found that an exogenous intervention in mortgage markets increases total consumption in the economy. When looking at the households based on their housing tenure - we found that households with the mortgage significantly increase their consumption expenditure, followed by a positive (but insignificant) increase in consumption expenditure of renters. The policy intervention has the smallest (and insignificant) increase in consumption expenditure for outright homeowners. In terms of age effects, we found a positive effect on young and middle-aged households but no effect on old households. Moreover, the interaction between housing tenure and age suggests most of the consumption response is driven by young and middle-aged mortgagors. We still find some positive effects for middle-aged renters and homeowners, but these are statistically insignificant.[15]

We identify the same groups of households in the model. We start with the total consumption, followed by housing tenure classification, age classification, and interaction between the two latter categories. We identify the households by tenure groups in the following way: renters (either renters that choose to rent or homeowners that sell their house), mortgagors (either mortgagors who make payments towards positive mortgage balance or refinancers), and outright owners (households that own the house and have zero mortgage outstanding). In terms of age - we reproduce the same mapping as in the empirical section: young household between the ages of 25 and 34, middle-aged households between ages 35 and 64, and older households above the age of 65.

We start by looking at the total consumption response of consumption to interest rate and mortgage rate shocks. As Fig. 8 illustrates, total expenditure increases almost immediately by around 0.01% percent, slowly decaying towards the initial steady-state value. The shape of the response in consumption is consistent with the shape of the response we found empirically: households increase their expenditure following the increase in agency mortgage purchases, reverting back to the old value as the shock decays (in our case, this decay is aligned with decay in interest and mortgage rate drop).[16]

We next divide the households into three groups based on the tenure housing status: mortgagors, renters, and outright homeowners. Fig. 9 plots the response of consumption of mortgagors (black solid line), renters (blue dotted line), and outright homeowners (red dashed line). As Figure illustrates, in line with the empirical finding, the positive response is total expenditure is driven by the positive response in expenditure by mortgagors. Contrary to the data, however, this positive response is out-weighted by the initial negative response in the expenditure of renters. Outright homeowners barely respond to changes in interest rates, similar to the data. We analyze the discrepancy between the response for renters in Section 5.3.

Next, we analyze the response of consumption by dividing the households into three age categories: young households (between ages 25 and 34), middle-aged households (between ages 35 and 65), and older households (above age 65). Fig. 9 plots the response of consumption for these groups: young (black solid line), middle-aged (blue dotted line), and older (red dashed line) households. Along the age dimension, the responses of consumption are consistent with those found in the data: we find a large response in simulated expenditure for young and middle-aged households and a smaller response for older households. Again, we leave the explanation of the transmission mechanism to Section 5.3.

5.3. Understanding transmission mechanism

We next analyze what is the transmission mechanism that policy operates through and what drives the increase in consumption reported above (both the overall and the heterogeneous response). We break down the discussion into several

[15] See Online Appendix.

[16] Numerically, the model does not reproduce the magnitude of the response found in the data. There are two possible reasons for this. First, we feed two shocks to the model: the shock to the interest rates and shock to the mortgage rates; in reality, there might be other shocks that accompany the increase in agency mortgage purchases - change in house prices, that we analyze in Online Appendix F, changes in preferences over housing (as analyzed, for example, in Kaplan et al. (2020), among others. Second, we do not estimate and match the marginal propensity to consume to interest rate shocks of the model to that of the data. As such, the magnitude of any changes of consumption to interest rate shocks could be under- or over-estimated. Instead, when analyzing the results, we rather focus on the shape of the response and the possible transmission mechanisms.

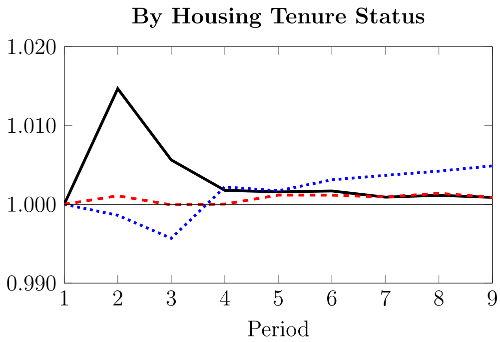

Fig. 9. The figure plots the simulated path of average expenditure for households based on the housing tenure status. Black solid line is for mortgagors, blue dashed line is for renters, and red dotted line is for outright homeowners. Initial period value is normalized to 1.

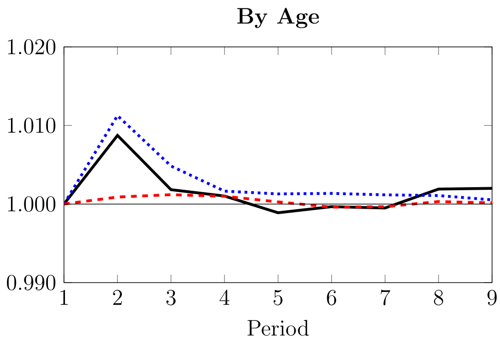

Fig. 10. The figure plots the simulated path of average expenditure for households based on their age. Black solid line is for young, blue dashed line is for middle-aged, and red dotted line is for older households. Initial period value is normalized to 1.

parts. We will start by analyzing what drives the increase in expenditure among mortgagors and how can the structural model shed light on that. We will then try to understand the discrepancy in the response of renters between the empirical finding and the model. Finally, we will look at the age dimension: why do young and middle-aged increase their expenditure the most?

Mortgagors As documented in Figs. 8 and 9, following a decrease in both the interest rates and the mortgage rates, households increase their expenditure, and this increase is mostly driven by the increase in the consumption of mortgagors (the former is fully consistent while the latter is partly consistent with empirical findings).

In the model, mortgage rates are important from two different perspectives for households taking out the mortgage: the mortgage rate determines the initial minimum payment requirement (payment to income constraint), and they determine the size of all future payments the households have to make. Which of those effects benefits households the most is not obvious apriori. The lower future payments on the debt will benefit all households with an outstanding mortgage balance. Relaxation of the payment to income constraint is not as straightforward. On the one hand, a lower mortgage rate implies lower minimum payment requirement, and as such, the payment to income constraint is less likely to bind (see equations (10) and (11)). On the other hand, when deciding to refinance, households still need to pay the fixed cost of taking out a mortgage. As such, it becomes a quantitative question: the size of the drop in mortgage rates (and how it translates into both lower mortgage payments and relaxation of the PTI constraint) will determine a threshold for households that will decide whether to refinance or not in the current period. Luckily, our structural model allows us to see exactly whether households refinance and if they do - what is their consumption level.

We start by looking at the dynamics of the refinancing following the shock. As Fig. 11 illustrates, following the drop in the interest and mortgage rates, the share of households that decided to refinance more than doubles (an increase of 236%). As households can anticipate the increase in the rates that followed the initial drop, two periods after the shock share of households that refinance drops below the steady-state value, slowly reverting to the initial value as the effect of the shock disappears.

Fig. 12 plots the percentage change of consumption for two different types of mortgagors - those that decide to refinance in a given period (black solid line) and those that continue with the payments (blue dotted line). As Figure illustrates, the households that decide to refinance increase their consumption more that the “average” mortgagor documented above. As the decrease in interest rates is temporary, the effect on consumption is also temporary (households who refinance also benefit longer from the decrease in mortgage and interest rates).

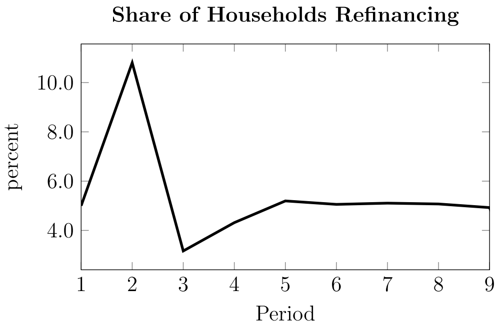

Fig. 11. The figure plots the implied share of household refinancing in the current period.

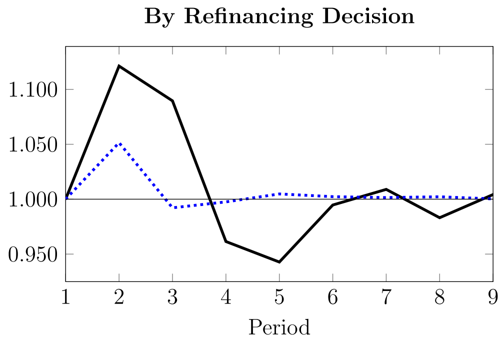

Fig. 12. The figure plots the simulated path of average expenditure for households based on the refinancing decision. Black solid line is for households that refinance, blue dashed line is for those that don't. Initial period is normalized to 1.

The results above point to the fact that access to refinancing option is an important propagation mechanism of mortgage rate changes. Moreover, the relaxation in the PTI constraint allows more households to benefit from the lower mortgage rates. Indeed, in the model, in the period following the shock, the minimum payment for households that decided to refinance in that period drops by around 20% (this is a combination of both the lower payments and smaller mortgage that households can take out; for first-time buyers the average minimum payment drops by 2% only), compared to the period before. This is relative to around 8% average drop in the payments that mortgagors make towards the debt.

Renters As documented above, following the decrease in both mortgage and interest rates, renters decrease initially decrease their expenditures following the drop in the interest and mortgage rates. Once the effect of the shock is over (after period 5), this group of households increases their expenditure slightly. Several important modeling assumptions can explain this initial decrease. In a very simple consumption-savings model, a drop in the interest rates would tell households to move away from savings and towards consumption. In the current setting, however, when solving the dynamic problem, households need to take into account the future value and possibility of being a homeowner. In the model, households have a preference of owning over renting (as dictated by the parameter ω), the parameter that is in principle, unobservable in the data. Moreover, on the one hand, becoming a homeowner becomes cheaper as the mortgage rates drop. On the other hand, in the model, households start with zero assets and need to accumulate assets before they can buy the house. As such, a drop in the interest rates actually hurts those households (taking into account the dynamic problem). Indeed, simulations indicate that preference for future housing dominates that of simple consumption/savings decisions, and renters maintain the level of savings in the period following a decrease in the interest rates. See also below for a breakdown by age and housing tenure status.[17]

Age Finally, we analyze the changes in the expenditure in the model based on age. As Fig. 10 indicates, young and middle-aged households drive the overall increase in expenditures, the result is also consistent with our empirical findings. To understand why young and middle-aged households respond most to expenditure changes, we can analyze life-cycle profiles in Fig. 6. Given the strong life-cycle pattern of homeownership, mortgage holdings, as well as indebtedness, both young and middle-aged households benefit the most from the decrease in mortgage payments.

Indeed, as top panel of Fig. 13 indicates, among younger households, mortgagors (solid black line) are the ones that increase the expenditure following the shock, while young renters decrease their expenditure (dotted blue line, see an

[17] Of course, since households also face uninsurable income risk, the precautionary savings motive also plays a role. The same is true for preference over bequest at the last period of life.

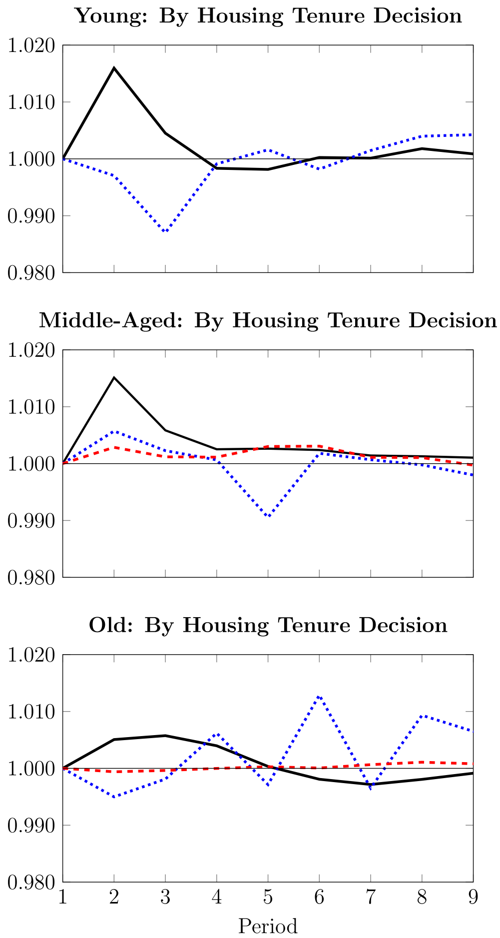

Fig. 13. The figure plots the simulated path of average expenditure for households based on age and hosing tenure status. Top panel is for younger households, middle panel is for middle-aged and bottom panel is for older. Black solid line represents mortgagors, blue dotted line - renters, and red dashed - outright owners. Initial period is normalized to 1.

argument as to why above). There are no outright homeowners among young households in our simulation. Similarly, among the middle-aged households (middle panel) most of the increase in expenditure is driven by mortgagors (black solid line), followed by middle-aged renters (blue dotted line). The fact that middle-aged renters increase their expenditure following the shock is in line with the argument we presented above: middle-aged renters are not as constrained in terms of assets as their younger counterparts and, as such, are more likely to switch from savings to expenditure following a decrease in the interest rates. Middle-aged owners (red dashed line) show a very mild increase. Finally, we find similar behavior among the older households (lower panel). Given the relatively small share of renters among the older households, the simulated path of expenditure for these households is quite noisy.

In Online Appendix we analyze an alternative experiment where we allow the house prices to change as a result of increase in agency mortgage purchases. In Online Appendix G we analyze a scenario where we don't allow for an "accommodative" monetary policy - that is, we keep the interest rates fixed and only allow for spread between two rate to change.

6. Conclusions

We study the heterogeneous impact of expansionary credit policies by combining exogenous policy changes in US federal housing agencies' mortgage holdings with household-level data from the Consumer Expenditure Survey and the Survey of Consumer Finances. We show that following an increase in agency purchases, households increase their consumption significantly. We then analyze the differences in consumption responses along two important dimensions: housing tenure status and age. We group households into three pseudo-cohorts based on their housing tenure status: renters, mortgagors, and homeowners. We show that following an increase in agency purchases, households with mortgages increase their spending, while outright homeowners and renters do not adjust their expenditures significantly. We also find that in terms of age dimension, most of the increase in consumption is driven by young and middle-aged households.

We explain this evidence through the lens of a Huggett (1996) type of heterogeneous life-cycle model with endogenous housing choice and idiosyncratic income risk. We estimate the structural model to match the aggregate and life-cycle moments of the data from the 2001 wave of the Survey of Consumer Finances, and we calibrate the mortgage market intervention to be consistent with empirical macroeconomic evidence. We show that lower mortgage rates, as well as the change in the spread between the rates, generate mortgage refinancing motives and explain the high observed increase in expenditure for mortgagors. We show that a low marginal propensity to consume and bequest motive outweighs the effect of a lower interest rate for outright homeowners. In the model, we find that renters actually decrease their expenditure following a cut in mortgage and interest rates, and we show that a combination of precautionary savings motive as well as future housing tenure status change can explain this fact in the model. Finally, we show that a combination of age-based and tenure-based responses goes in line with the evidence on the presence of wealthy hand-to-mouth households.

Data availability

I have sent the codes and the final dataset with Christian Zimmermann. Those include all files to replicate empirical and theoretical part of the paper.

Appendix A. Supplementary material

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.red.2022.11.005.

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